Lista 1.

1. Seja y um variável aleatória com distribuição cuja função de probabilidade é dada por

$$f(y;\mu) = \frac{4(y+1)\mu^y}{(\mu+2)^{y+2}}, \quad \mu > 0, \quad y = 0, 1, 2, \ldots$$

a) Mostre que a distribuição de y pertence à família exponencial linear

$$f(y;\mu) = \frac{4(y+1)\mu^y}{(\mu+2)^{y+2}} =$$

$$= \exp\{\log 4 + \log(y+1) + y\log\mu - (y+2)\log(\mu+2)\}$$

$$= \exp\{y\log\mu - y\log(\mu+2) - 2\log\mu+2 + \log 4 + \log(y+1)\}$$

$$= \exp\left\{y\log\left(\frac{\mu}{\mu+2}\right) - 2\log\mu+2 + \log 4 + \log(y+1)\right\}$$

$\Rightarrow \phi = 1 \quad \theta = \log\frac{\mu}{\mu+2} \quad b(\theta) = 2\log\mu+2 \quad c(y,\phi) = \log 4 + \log(y+1)$

$$e^\theta = \frac{\mu}{\mu+2} \Rightarrow e^\theta(\mu+2) = \mu \Rightarrow e^\theta\mu + 2e^\theta = \mu \Rightarrow \frac{2e^\theta}{1-e^\theta} = \mu$$

$$\Rightarrow b(\theta) = 2\cdot\log\left(\frac{2e^{\theta}}{1-e^{\theta}} + \frac{2(1-e^{\theta})}{(1-e^{\theta})}\right) = 2\cdot\log\frac{2}{1-e^{\theta}} = 2\log 2 - 2\log(1-e^{\theta})$$

Assim podemos reescrever $f(y;\mu)$ da forma

$$f(y;\mu) = \exp\left\{\phi[\theta y - b(\theta)] + c(y,\phi)\right\}$$

Logo a dist. de y pertence à família exponencial linear.

b) Encontre $E(y)$ e $Var(y)$

Sabemos que $E(y) = b'(\theta)$ e $Var(y) = b''(\theta)\phi$

e $b(\theta) = 2\log 2 - 2\log(1-e^{\theta}) \Rightarrow \frac{db(\theta)}{d\theta} = \frac{2e^{\theta}}{1-e^{\theta}}$

como $e^{\theta} = \frac{\mu}{\mu+2} \Rightarrow b'(\mu) = \frac{2\left(\frac{\mu}{\mu+2}\right)}{1-\left(\frac{\mu}{\mu+2}\right)} = \frac{2\left(\mu/\mu+2\right)}{2/\mu+2} = \mu$

* $1 = \frac{\mu+2}{\mu+2}$

$\Rightarrow$ $E(y) = \mu$

$$b''(\theta) = \frac{d^2 b(\theta)}{d\theta^2} = \frac{d\left(\frac{2e^{\theta}}{1-e^{\theta}}\right)}{d\theta} = \frac{2e^{\theta}(1-e^{\theta}) + 2e^{\theta}e^{\theta}}{(1-e^{\theta})^2}$$

$$= \frac{2e^{\theta} - 2e^{2\theta} + 2e^{2\theta}}{(1-e^{\theta})^2} = 2e^{\theta}/(1-e^{\theta})^2$$

* $e^{\theta} = \mu/\mu+2$

$$= 2\cdot\left(\frac{\mu}{\mu+2}\right)\cdot\frac{1}{\left(1-\frac{\mu}{\mu+2}\right)^2} = \frac{2\mu}{\mu+2}\left(\frac{1}{\frac{2}{\mu+2}}\right)^2$$

$$= \frac{2\mu}{\mu+2}\cdot\left(\frac{\mu+2}{2}\right)^2 = \frac{\mu(\mu+2)}{2} \Rightarrow \boxed{Var(y) = \frac{\mu(\mu+2)}{2}}$$

c) Função Desvio. a.a de tamanho n. μ_i

Sabemos que $D(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}}) = 2\phi\left\{\sum\left[y_i(\tilde{\theta}_i - \hat{\theta}_i) + b(\hat{\theta}_i) - b(\tilde{\theta}_i)\right]\right\}$

$\tilde{\mu}_i = y_i$

$\tilde{\theta}_i = \log\left(\frac{y_i}{y_i+2}\right)$, $\hat{\theta}_i = \log\left(\frac{\hat{\mu}_i}{\hat{\mu}_i+2}\right)$ $b(\theta) = 2\log 2 - 2\log(1-e^{\theta})$

$$\Rightarrow D(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}}) = 2\phi\left\{\sum\left[y_i\left(\log\left(\frac{y_i}{y_i+2}\right) - \log\left(\frac{\hat{\mu}_i}{\hat{\mu}_i+2}\right) + b\left(\log\frac{\hat{\mu}_i}{\hat{\mu}_i+2}\right) - b\left(\log\frac{y_i}{y_i+2}\right)\right]\right\}$$

$$= 2\sum\left[y_i\left(\log y_i - \log\left(\frac{\hat{\mu}_i}{\hat{\mu}_i+2}\right) + 2\log 2 - 2\log\left(1-\frac{\hat{\mu}_i}{\hat{\mu}_i+2}\right) - 2\log 2 + 2\log\left(1-\frac{y_i}{y_i+2}\right)\right)\right]$$

2. Seja Y uma v.a com dist. Bernoulli

a) Mostre que a dist. $y \in$ FEL $\quad y \in \{0,1\}$

$$Y \sim Bern(p) \Rightarrow f(y;p) = p^y \cdot (1-p)^{1-y} = \exp\{y \log p + (1-y)\log(1-p)\}$$

$$= \exp\{y \log p + \log 1-p - y \log 1-p\} = \exp\{y(\log p - \log 1-p) + \log 1-p\}$$

$$= \exp\{y(\log \tfrac{p}{1-p}) + \log 1-p\} \qquad \theta = \log \frac{p}{1-p} \Rightarrow e^\theta = \frac{p}{1-p} \Rightarrow e^\theta - pe^\theta = p$$

$$\Rightarrow e^\theta = p(1+e^\theta) \Rightarrow p = \frac{e^\theta}{1+e^\theta}$$

$$-b(\theta) = \log 1-p = \log 1 - \frac{e^\theta}{1+e^\theta}$$

$$= \log \frac{1+e^\theta - e^\theta}{1+e^\theta} = \log \frac{1}{1+e^\theta} = -\log 1+e^\theta$$

$$\Rightarrow b(\theta) = \log 1+e^\theta \qquad \phi = 1 \quad c(y,\phi) = 0$$

Como podemos escrever $f(y,p)$ da forma $\exp\{\phi(\theta y - b(\theta)) + c(y,\phi)\}$

com $\phi = 1$, $\theta = \log \frac{p}{1-p}$, $b(\theta) = \log 1+e^\theta$ $c(y,\phi) = 0$, a dist. de y pertence de fato a FEL.

b) $E(y)$ e $Var(y)$ = ?

Sabemos que $E(y) = b'(\theta)$ e $Var(y) = b''(\theta)\cdot\phi$

e $b(\theta) = \log 1+e^{\theta} \Rightarrow \frac{d\,b(\theta)}{d\theta} = \frac{1}{1+e^{\theta}}\cdot e^{\theta} = \frac{e^{\theta}}{1+e^{\theta}} = p$

$\Rightarrow$ $E(y) = p$

$$b''(\theta) = \frac{d\,\frac{e^{\theta}}{1+e^{\theta}}}{d\theta} = \frac{e^{\theta}(1+e^{\theta}) - e^{\theta}\cdot e^{\theta}}{(1+e^{\theta})^2} = \frac{e^{\theta}+e^{2\theta}-e^{2\theta}}{(1+e^{\theta})^2} = \frac{e^{\theta}}{(1+e^{\theta})^2}$$

$$= \frac{e^{\theta}}{1+e^{\theta}}\cdot\frac{1}{1+e^{\theta}} = \frac{e^{\theta}}{1+e^{\theta}}\cdot\left(\frac{1+e^{\theta}-e^{\theta}}{1+e^{\theta}}\right) = p\cdot(1-p)$$

$\Rightarrow$ $Var(Y) = p(1-p)$

c) função desvio, supondo uma amostra de n v.a's iid, com μ_i de parâmetro

Sabemos que $D(y,\mu) = 2\phi\left\{\Sigma\left[y_i(\tilde{\theta}_i - \hat{\theta}_i) + b(\hat{\theta}_i) - b(\tilde{\theta}_i)\right]\right\}$

$\tilde{\theta}_i = \log\frac{y_i}{1-y_i}$, $\hat{\theta}_i = \log\frac{\hat{\mu}_i}{1-\hat{\mu}_i}$ $b(\theta) = \log 1+e^{\theta}$

$$\Rightarrow D(\underset{\sim}{y},\underset{\sim}{\mu}) = 2\cdot\Sigma\left[y_i\left(\log\frac{y_i}{1-y_i} - \log\frac{\hat{\mu}_i}{1-\hat{\mu}_i}\right) + \log\left(1+\frac{\hat{\mu}_i}{1-\hat{\mu}_i}\right) - \log\left(1+\frac{y_i}{1-y_i}\right)\right]$$

3. ex 6 do cap 2:

Mostre que a distribuição logarítmica com função de probabilidade

$$f(y;p) = \frac{p^y}{-y \log 1-p}, \quad y = 1, 2, \cdots, \text{ e } 0 < p < 1$$

pertence a família exponencial. Calcule μ e $V(\mu)$. Função de desvio supondo amostra de tamanho n. $R^2 = ?$

$$f(y;p) = \frac{e^y}{-y \log 1-p} = \exp\{y \log p - \log(y) - \log(-\log 1-p)\}$$

$$= \exp\{y \log p - \log -\log 1-p - \log y\}$$

$$\Rightarrow \theta = \log p \quad b(\theta) = \log \log 1-p \quad c(y,\phi) = -\log y$$

* $p \in (0,1) \Rightarrow \theta < 0$

$$\Rightarrow p = e^\theta \quad b(\theta) = \log \log 1 - e^\theta \quad \phi = 1$$

Como a densidade $f(y;p)$ pode ser escrita da forma

$f(y,p) = \exp\{\phi(\theta y - b(\theta) + c(y,\phi)\}$, então a dist. logarítmica pertence a fam. exponencial.

Sabendo que $\mu = E(y) = b'(\theta)$, e $V(\mu) = b''(\theta) = Var(y)$

$b(\theta) = \log\log 1 - e^{\theta} \Rightarrow b'(\theta) = \frac{1}{\log 1 - e^{\theta}} \cdot \frac{1}{1 - e^{\theta}} \cdot -e^{\theta}$

$* e^{\theta} = \rho$

$\Rightarrow b'(\theta) = \frac{-\rho}{\log(1-\rho)\cdot(1-\rho)} = \mu$

$$b''(\theta) = \frac{d}{d\theta}\left(\frac{-e^{\theta}}{\log 1 - e^{\theta}} \cdot \frac{1}{(1-e^{\theta})}\right) = \frac{-e^{\theta}\cdot[\log(1-e^{\theta})\cdot(1-e^{\theta})] - (e^{\theta})\cdot\left[\frac{1\cdot e^{\theta}}{1-e^{\theta}}\cdot(1-e^{\theta}) + \log(1-e^{\theta})\cdot(e^{\theta})\right]}{[\log(1-e^{\theta})\cdot(1-e^{\theta})]^2}$$

$$= \frac{-e^{\theta}\log(1-e^{\theta})\ (1-e^{\theta}) + e^{\theta}\cdot\left[-e^{\theta} + \log(1-e^{\theta}) - e^{\theta}\right]}{[\log(1-e^{\theta})\cdot(1-e^{\theta})]^2}$$

$$= \frac{-\rho\log(1-\rho)(1-\rho) + \rho[-\rho + \log(1-\rho)(-\rho)]}{[\log(1-\rho)(1-\rho)]^2}$$

$$= \frac{-\rho\log(1-\rho) + \rho^2\log 1-\rho - \rho^2 - \rho^2\log(1-\rho)}{[\log(1-\rho)(1-\rho)]^2}$$

$$= [-\rho^2 - \rho\log(1-\rho)] \ / \ [\log(1-\rho)(1-\rho)]^2$$

Função desvio: $D(\underset{\sim}{y}, \underset{\sim}{\mu}) = 2\phi\Sigma[y_i(\tilde{\theta}_i - \hat{\theta}_i) + b(\hat{\theta}_i) - b(\tilde{\theta}_i)]\}$

* $e^{\theta} = \rho \Rightarrow \theta = \log \rho \quad \tilde{\rho} = y_i$?

$D(\underset{\sim}{y}, \hat{\rho}) = 2 \cdot \Sigma[y(\log y_i$

4. Considere MLG com resposta Bernoulli e função de ligação log log.

$\Rightarrow g(\mu_i) = -\log(-\log(\mu_i))$

a) μ_i em função de η_i

$\eta_i = g(\mu_i) = -\log(-\log(\mu_i)) \Rightarrow e^{-\eta_i} = -\log \mu_i$

$\Rightarrow e^{-e^{-\eta_i}} = \mu_i$

$\mu_i = e^{-e^{-\eta_i}}$

b) Mostre que $0 < \tilde{\mu}_i < 1$

$\hat{\mu}_i = e^{-e^{-\hat{\eta}_i}} = e^{-e^{-x_i\hat{\beta}}}$, note que $e^{k} > 0, \forall k \in \mathbb{R}$

sabemos que $e^{k} < 1 \Leftrightarrow k < 0$, note que $e^{-\hat{\eta}_i} > 0$

$\Rightarrow -e^{-\hat{\eta}_i} < 0$, então $e^{-e^{-\hat{\eta}_i}} < 1 \Rightarrow$

$0 < e^{-e^{-\hat{\eta}_i}} < 1$

5. Considere MLG com resposta Gama. Obtenha função de μ_i, vetor z e a matriz W. P estimar $\underset{\sim}{\beta}$ por max. verossim.

a) f de ligação canônica b) f de ligação log. $\Rightarrow g(\mu_i) = \log \mu_i$

a) p/ Gama, a função de ligação canônica é $\eta_i = g(\mu_i) = \frac{1}{\mu_i} \Rightarrow g'(\mu_i) = \frac{-1}{\mu_i^2}$

e $V(\mu_i) = \mu_i^2$ como $\underset{\sim}{z} = \eta + G(y - \underset{\sim}{\mu})$ e $w_i = \frac{1}{V_i}\left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2$

* $G = \text{diag}\{g'(\mu_1) \cdots g'(\mu_n)\}$

* $\frac{\partial \mu_i}{\partial \eta_i} = \left(\frac{\partial \eta_i}{\partial \mu_i}\right)^{-1} = \left(g'(\mu_i)\right)^{-1} = \left(\frac{-1}{\mu_i^2}\right)^{-1} = (-\mu_i^2) \Rightarrow w_i = \frac{1}{\mu_i^2}(-\mu_i^2)^2 = \mu_i^2$

e $z_i = \eta_i + (y_i - \mu_i)\cdot \frac{d\eta_i}{d\mu_i} = \frac{1}{\mu_i} - (y_i - \mu_i)\cdot\frac{1}{\mu_i^2} = \frac{1}{\mu_i} - \frac{y_i}{\mu_i^2} + \frac{\mu_i}{\mu_i^2} = \frac{2}{\mu_i} - \frac{y_i}{\mu_i^2}$

$\Rightarrow W = \text{diag}\{\mu_1^2, \cdots, \mu_n^2\}$ e $\underset{\sim}{z} = \begin{pmatrix} \frac{2}{\mu_1} - \frac{y_1}{\mu_1^2} \\ \vdots \\ \frac{2}{\mu_n} - \frac{y_n}{\mu_n^2} \end{pmatrix}$

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b) $g(\mu_i) = \log \mu_i \Rightarrow g'(\mu_i) = 1/\mu_i$ $V(\mu_i) = \mu_i^2$

$W = \frac{1}{V_i}\left(g'(\mu_i)^{-1}\right)^2 = \frac{1}{\mu_i^2}\cdot(\mu_i)^2 = 1 \Rightarrow W = I_{n\times n}$

$$z_i = \eta_i + (y_i - \mu_i)\, g'(\mu_i) = \log \mu_i + (y_i - \mu_i) \cdot \frac{1}{\mu_i} = \log \mu_i + \frac{y_i}{\mu_i} - 1$$

$$\Rightarrow \underset{\sim}{z} = \begin{bmatrix} \log \mu_1 + \frac{y_1}{\mu_1} - 1 \\ \vdots \\ \log \mu_n + \frac{y_n}{\mu_n} - 1 \end{bmatrix}_{n \times 1}$$

6. Mostre que no MLG com Y normal, o estimador de ϕ por MV é $\frac{n}{D^*(y, \hat{\mu})}$

se Y Normal com $Var(y_i) = \phi = \frac{1}{\sigma^2}$ como parametro de precisão $\Rightarrow y_i \sim N(\mu_i, \phi^{-1})$

temos $L(\underset{\sim}{\mu}, \phi; \underset{\sim}{y}) = \prod_i^n \frac{\phi}{\sqrt{2\pi}} \cdot \exp\left\{\frac{-\phi}{2}(y_i - \mu_i)^2\right\} = \left(\frac{\phi}{2\pi}\right)^{\frac{n}{2}} \cdot \exp\left\{\frac{-\phi}{2} \cdot \Sigma (y_i - \mu_i)^2\right\}$

$$\Rightarrow \log(L(\underset{\sim}{\mu}, \phi; \underset{\sim}{y})) = l(\underset{\sim}{\mu}, \phi; \underset{\sim}{y}) = \frac{n}{2} \cdot \log\left(\frac{\phi}{2\pi}\right) - \frac{\phi}{2} \cdot \Sigma (y_i - \mu_i)^2$$

para $\sigma^2 > 0$ ou seja $\phi > 0$, podemos fazer $\frac{dl}{d\phi} = \frac{n}{2} \frac{1}{\phi} - \frac{1}{2} \sum_{i=1}^{n} (y_i - \mu_i)^2$

e se $\left.\frac{d\ell}{d\phi}\right|_{\phi=\hat{\phi}} = 0 \Rightarrow \frac{n}{2\hat{\phi}} = \frac{\sum (y_i-\mu_i)^2}{2} \Rightarrow \hat{\phi} = \frac{n}{\sum (y_i-\mu_i)^2}$

Como $\frac{d^2\ell}{d\phi^2} = \frac{-n}{2\phi^2} < 0$, de fato a função ℓ tem um ponto max em $\hat{\phi}$.

Agora note que no modelo saturado $\tilde{\mu}_i = y_i$ e $\hat{\theta}_i = \hat{\mu}_i$ $\quad^{*}\theta_i = \mu_i$ em Y/Normal

$$\Rightarrow D^*(y;\hat{\mu}) = 2\cdot\sum\left\{ y_i(y_i-\hat{\mu}_i) + \frac{\hat{\mu}_i^2}{2} - \frac{y_i^2}{2} \right\}$$

$$= 2\cdot\sum \frac{\hat{\mu}_i^2}{2} + \frac{y_i^2}{2} - y_i\hat{\mu}_i = \sum (\hat{\mu}_i^2 - 2\hat{\mu}_i y_i + y_i^2) = \sum (y_i-\hat{\mu}_i)^2$$

Logo o estimador de ϕ de fato é $\hat{\phi} = \frac{n}{\sum (y_i-\hat{\mu}_i)^2} = \frac{n}{D^*(y_i,\hat{\mu}_i)}$

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7. MLG, resposta gaussiana inversa, com função de ligação canônica

$\Rightarrow g(\mu) = \frac{1}{\mu^2} = \eta$. e apenas uma única variável preditora

e temos dados: $\underset{\sim}{y} = [2{,}3 \quad 1{,}8 \quad 1{,}5 \quad 1{,}3 \quad 1{,}2]^T$

$\underset{\sim}{x} = [2 \quad 4 \quad 6 \quad 9 \quad 10]^T$

Estime β_0 e β_1 por MV, usando manualmente métodos de min.quad.

Estime β_0 e β_1 por método dos momentos, e por MV.

$V(\mu) = \mu_i^3 \qquad \eta_i = x_i^T \beta = \beta_0 + \beta_1 x_i = \frac{1}{\mu_i^2} \Rightarrow \mu_i = (\beta_0 + \beta_1 x_i)^{-\frac{1}{2}}$

$g(\mu) = \mu^{-2}$

$g'(\mu_i) = -2\mu^{-3} \Rightarrow w_i = \frac{1}{V_i}\left(g'(\mu_i)\right)^{-1} = \frac{1}{\mu_i^3} \cdot \left(\frac{-1}{2}\mu^3\right)^2 = \frac{\mu_i^3}{4}$

$z_i = \eta_i + (y_i - \mu_i) \cdot g'(\mu_i) = \frac{1}{\mu_i^2} + (y_i - \mu_i) \cdot \frac{-2}{\mu_i^3} = \frac{1}{\mu_i^2} - \frac{2(y_i - \mu_i)}{\mu_i^3}$

Inicialização: tomando $\mu_i^{(0)} = y_i \Rightarrow W^{(0)} = \text{diag}\left\{\frac{y_1^3}{4}, \frac{y_2^3}{4}, \cdots, \frac{y_5^3}{4}\right\}$

e $z_i^{(0)} = \frac{1}{y_i^2} \Rightarrow \underset{\sim}{\beta}^{(1)} = \underset{2\times5\ \ 5\times5\ \ 5\times2}{\left(X^T W^{(0)} X\right)^{-1}} \underset{2\times5\ \ 5\times5\ \ 5\times1}{X^T W^{(0)} z^{(0)}}$

$$\beta^{(1)} = \left(\underset{2\times5}{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 9 & 10 \end{bmatrix}} \underset{5\times5}{\begin{bmatrix} \frac{2,3^3}{4} & & & & \\ & \frac{1,8^3}{4} & & & \\ & & \frac{1,5^3}{4} & & \\ & & & \frac{1,3^3}{4} & \\ & & & & \frac{1,2^3}{4} \end{bmatrix}} \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 9 \\ 1 & 10 \end{bmatrix} \right)^{-1}$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 9 \\ 1 & 10 \end{bmatrix}^T \begin{bmatrix} \frac{2,3^3}{4} & & & & \\ & \frac{1,8^3}{4} & & & \\ & & \frac{1,5^3}{4} & & \\ & & & \frac{1,3^3}{4} & \\ & & & & \frac{1,2^3}{4} \end{bmatrix} \begin{bmatrix} \frac{1}{2,3^2} & & & & \\ & \frac{1}{1,8^2} & & & \\ & & \frac{1}{1,5^2} & & \\ & & & \frac{1}{1,3^2} & \\ & & & & \frac{1}{1,2^2} \end{bmatrix}$$

* $W = \text{diag}\{3,04175 ; 1,45800 ; 0,84375 ; 0,54925 ; 0,432\}$

$\Rightarrow X^T W^{(0)} X = \begin{bmatrix} 6,32475 & 26,24125 \\ 26,24125 & 153,55925 \end{bmatrix}$

$\Rightarrow (X^T W^{(0)} X)^{-1} = \begin{bmatrix} 0,5433 & -0,092 \\ -0,092 & 0,123 \end{bmatrix}$

$(X^T W^{(0)} \underset{\sim}{z}) = \begin{bmatrix} 2,025 \\ 11,125 \end{bmatrix} \quad \Rightarrow \beta^{(1)} = \begin{bmatrix} 0,067 \\ 0,067 \end{bmatrix}$

feito a 1ª interação, os proximos passos serão:

$\eta^{(1)} = X^T \beta^{(1)}$ e $\mu^{(1)} = (q^{(1)})^{\frac{-1}{2}} \Rightarrow \mu^{(1)} = (X^T \beta^{(1)})^{\frac{-1}{2}}$

$\Rightarrow z^{(1)} = [0,19 \quad 0,31 \quad 0,44 \quad 0,59 \quad 0,69]^T$

$W^{(1)} = \text{diag}\{3,037 ; 1,44 ; 0,87 ; 0,52 ; 0,45\}$

e $\beta^{(2)} = (X^T W^{(1)} X)^{-1} X^T W^{(1)} z^{(1)}$ e $X^T W^{(1)} X = \begin{bmatrix} 6,32 & 26,25 \\ 26,25 & 153,6 \end{bmatrix}$

$X^T W^{(1)} z^{(1)} = \begin{bmatrix} 2,026 \\ 11,134 \end{bmatrix} \Rightarrow \beta^{(2)} = \begin{bmatrix} 0,067 \\ 0,067 \end{bmatrix}$ parece que convergiu já

Estimação de ϕ por metodo dos momentos:

Temos $V(\mu) = \mu_i^3$ e $Var(y) = \frac{1}{\phi} \cdot V(\mu) \Rightarrow Var(y) = \phi \cdot \mu_i^3$

$$\Rightarrow E((y_i-\mu_i)^2) = \frac{1}{\phi}\mu_i^3 \Rightarrow E\left(\frac{(y_i-\mu_i)^2}{\mu_i^3}\right) = \frac{1}{\phi} \Rightarrow \frac{1}{\hat{\phi}} = \frac{1}{n}\cdot\sum\left[\frac{(y_i-\mu_i)^3}{\hat{\mu}_i^3}\right]$$

$$\Rightarrow \hat{\phi}_{mm} = \left[\sum_{i=1}^{5}\left(\frac{(y_i-\hat{\mu}_i)^2}{\hat{\mu}_i^3}\right)\cdot n\right]^{-1}$$

* usando $\hat{\mu}_i$ estimado da 1ª iteração

e $\hat{\mu}^{(1)} = [2{,}29;\ 1{,}79;\ 1{,}52;\ 1{,}27;\ 1{,}2]$

$$\Rightarrow \hat{\phi}_{mm} = \frac{5}{0{,}0005737} \simeq 8714{,}418$$

alternativa: Como temos $K=2$ parâmetros, podemos fazer uma correção com a dist assintotica pois

$$\phi\,\Sigma\left[\frac{(y_i-\hat{\mu}_i)^2}{V(\hat{\mu}_i)}\right] \longrightarrow \chi_{n-K-1}$$

$$\Rightarrow \hat{\phi}_{mm_2} = \frac{n-K-1}{\Sigma\frac{(y_i-\mu_i)^3}{\mu_i^3}} = 3485{,}767$$

Estimação de ϕ por MV:

$$\frac{\partial l(\beta,\phi)}{\partial \phi} = U_\phi = \sum\left(\frac{-y_i}{2\mu_i^2} + \frac{1}{\mu_i}\right) + \sum\left(\frac{1}{2\phi} - \frac{1}{2y_i}\right)$$

$$U_\phi(\hat{\phi}) = 0 \Rightarrow \frac{n}{2\hat{\phi}} = \sum\left[\frac{y_i}{2\mu_i^2} - \frac{1}{\mu_i} + \frac{1}{2y_i}\right] = \sum\left[\frac{y_i^2}{2\mu_i^2 y_i} - \frac{2\mu_i y_i}{2\mu_i^2 y_i} + \frac{\mu_i^2}{2y_i\mu_i^2}\right]$$

$$= \sum\frac{(y_i-\hat{\mu}_i)^2}{2y_i\mu_i^2}, \quad \frac{n}{2\hat{\phi}} = \sum\frac{(y_i-\hat{\mu}_i)^2}{2y_i\hat{\mu}_i^2} \Rightarrow \hat{\phi} = \frac{n}{\sum\left[\frac{(y_i-\hat{\mu}_i)^2}{y_i\hat{\mu}_i^2}\right]}$$

usando $\hat{\mu}_i = \hat{\mu}_i^{(1)}$, temos $\sum\left[\frac{(y_i-\hat{\mu}_i)^2}{y_i\hat{\mu}_i^2}\right] = 0,0005706346$

$$\hat{\phi} = \frac{5}{0,0005706346} \simeq 8762,775$$